Documentació servidor DNS

El primer pas que hem de fer per configurar el servidor DNS es instalar les eines del bind, per fer-ho hem de aplicar la seguent comanda

```
user@vm-server ~> sudo apt install bind9 bind9utils _
```

Per treballar millor anirem al directori bind

```
user@vm-server ~> cd /etc/bind/
user@vm-server /e/bind> _
```

I configurarem l'arxiu named.conf.local

```
user@vm-server /e/bind> sudo nano named.conf.local
```

Aqui dins escriurem el seguent per crear la zona pol.navarro.local

Aixi configurem la zona i preparem per connectar el slave a la zone

```
zone "pol.navarro.local"{
type master;
file "/etc/bind/db.pol.navarro.local";
allow-transfer {
192.168.24.20;
        };_
};
```

;

A continuacio crearem l'arxiu db.pol.navarro.local fent servir com a base l'arxiu db.local.

```
user@vm-server /e/bind> sudo cp db.local db.pol.navarro.local
user@vm-server /e/bind> ls
bind.keys  db.127  db.empty  db.pol.navarro.local  named.conf.default-zones  named.conf.options  zones.rfc1918
db.0       db.255  db.local  named.conf            named.conf.local          rndc.key
```

La primera configuracio que s'ha de fer en aquest document es configurar els noms de les zones, per aixo canviarem els que son necessaris per que la zona funcioni

I

```
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     pol.navarro.local. root.pol.navarro.local. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      ns.pol.navarro.local.
@       IN      A       ns2.pol.navarro.local.
```

despres configurarem les ips, els clients i els serveis,

(falta @ IN MX 1 MAIL)

```
;
@       IN      NS      ns.pol.navarro.local.
@       IN      A       ns2.pol.navarro.local.
@       IN      A       192.168.24.1
ns      IN      A       192.168.24.1
www     IN      CNAME   mail
ns2     In      A       192.168.24.2
ftp     IN      CNAME   mail
mail    IN      A       192.168.24.3
@       IN      MX      1       mail
w3      In      A       192.168.24.4
ftp3    IN      CNAME   w3
printer IN      A       192.168.24.50
pc1     IN      A       192.168.24.51
pc2     IN      A       192.168.24.52
```

Tornarem a editar l'arxiu named.conf.local i crearem una nova zona tambe amb allow-transfer

```
zone "test.local"{
type master;
file "/etc/bind/db.test.local";
allow-transfer {
192.168.24.20;
        };
};
```

Ara tornarem a crear un arxiu amb la base de db.local, que es dira db.test.local

```
user@vm-server /e/bind> sudo cp db.local db.test.local
```

I configurarem el seguent amb la configuracio necesaria, en aquest cas es nomes el mail i el www

```
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     test.local. root.test.local. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
```

Per comprobar la sintaxis dels documents farem les seguents comandes

```
user@vm-server /e/bind> named-checkconf named.conf.local
```

Per comprobar la sintaxis de les zones

```
user@vm-server /e/bind> named-checkconf db.pol.navarro.local
db.pol.navarro.local:1: syntax error near ';'
user@vm-server /e/bind [1]> [2~
```

Aqui podem veure un error aprop d'un punt i coma

Configuracio del netplan

```
user@vm-server /e/bind> sudo nano /etc/netplan/50-cloud-init.yaml
```

Aqui posarem la ip del nostre servidor per activar el netplan

```
# This file is generated from info
# to it will not persist across an
# network configuration capabiliti
# /etc/cloud/cloud.cfg.d/99-disabl
# network: {config: disabled}
network:
    ethernets:
        enp0s3:
            addresses:
              - 192.168.24.1/24
    version: 2
```

```
user@vm-server /e/bind> sudo netplan apply
```

A continuacio passarem a la configuracio del slave

Seguirem els passos inicials de la instal·lacio

```
user@vm-server ~> sudo apt install bind9 bind9utils
```

anirem al directori bind

```
user@vm-server ~ [1]> cd /etc/bind
```

Configurarem l'arxiu named.conf.local com a un slave

```
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "pol.navarro.local" {
type slave;
file "db.pol.navarro.local";
masters { 192.168.24.1; };
};

zone "test.local" {
type slave;
file "db.test.local";
masters { 192.168.24.1; };
};
```

Despres d'aixo configurarem el netplan del slave, la ip ha de ser la mateixa que vem posar con a ns2 en el master, en aquest cass 192.168.24.20

```
# This file is generated from information provided by the datasource.  Changes
# to it will not persist across an instance reboot.  To disable cloud-init's
# network configuration capabilities, write a file
# /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg with the following:
# network: {config: disabled}
network:
    ethernets:
        enp0s3:
            addresses:
              - 192.168.24.20/24
    version: 2
```

Per comprobar agafarem un client d'ubuntu i farem la comanda nslookup i posarem per exemple mail o ftp, que son els subdominis.

Inversa:

copiar document db.255 o db.192 i posar cada apartat (mail, ftp, w3…) amb PTR

exemple:

2 IN PTR NS2.pol.local

```
 GNU nano 7.2                                                  db.inversa
;
; BIND reverse data file for broadcast zone
;
$TTL    604800
@       IN      SOA     dani.local. root.dani.local. (
                              3         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      ns1.dani.local.
@       IN      NS      ns2.dani.local.
1       IN      PTR     ns1.dani.local.
2       IN      PTR     ns2.dani.local.
10      IN      PTR     mail.dani.local.
11      IN      PTR     ftp.dani.local.
12      IN      PTR     nazismo.dani.local.
10      IN      PTR     www.dani.local.
11      IN      PTR     w3.dani.local.
11      IN      PTR     navarro.dani.local.
12      IN      PTR     negros.dani.local.
12      IN      PTR     mierda.dani.local.
```

master:

Crear una zona inversa fent servir la ip del reves amb in-addr.arpa i els dígits de la ip al reves, la quantitat depen de la máscara

Exemple: 192.168.1.1/24

1.168.192.in-addr.arpa

172.16.1.1/16

16.172.in-addr.arpa



slave:

Amb l'slave fem el mateix que amb el master, amb el in-addr.arpa



Configuracio client

Per a fer les comprovacions accedir a /etc/resolv.conf

```
  GNU nano 7.2                          /etc/resolv.conf *
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 192.168.1.1
options edns0 trust-ad
search dani.local
```

S'ha de canviar aquest document i posar el nom del domini i la ip d'aquest

LOGS:

```
user@vm-server ~> sudo tail -f /var/log/syslog | grep named
```